

# DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology

Advised by [Prof. HyunJoo Oh](#) and [Prof. Sehoon Ha](#)

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## Research Interests

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My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

## Education

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**Georgia Tech, Ph.D. in Computer Science**

2022–Present

- [School of Interactive Computing](#); advised by Profs. [Oh](#) & [Ha](#)
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

**Korea University, M.Eng. in ECE**

2020–2022

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

**Korea University, B.Eng. in EE**

2012–2020

## Experience

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**Ph.D. Research Intern, Dolby Laboratories**

2024–2025

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

**MLOps Engineering Intern, LG Uplus Corp.**

2023

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

**Research Scientist, Protopia AI Inc.**

2021–2022

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

**Cloud Architect · AWS Educate Cloud Ambassador**

2020–2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

**Server Engineer, Flysher Inc.**

2019

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

**System Engineer, AKA Intelligence (Musio)**

2017–2019

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

## Selected Publications

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- 1 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KIPS Trans. Softw. & data*, Mar. 2023.